

# ASHWANI.DEVOPS

GitHub Pages + GoDaddy DNS + Custom Domain + HTTPS

Complete practical guide — what we wanted to build, every step we performed, why we did it, what each component does, and a real-world recruiter use case.

Final target: <https://ashwaniai.site>

## 1. Our Goal

We wanted a professional Azure DevOps Engineer portfolio on the existing domain [ashwaniai.site](https://ashwaniai.site), without buying GoDaddy Website Builder. The site showcases Azure, Terraform, Kubernetes/AKS, CI/CD, DevSecOps, monitoring/LGTM, projects and architecture.



## 2. Domain vs Hosting

GoDaddy manages the domain and DNS. GitHub stores the source code and GitHub Pages hosts the static website. DNS connects the domain to the hosting service.



## STEP 1 — GitHub Repository

We created `ashwanishahi155/ashwani-devops-portfolio`.

GitHub Repo\n |\n +-- index.html

Why / use case: Why: the repository is the version-controlled home for the website source code.

## STEP 2 — index.html

We created `index.html` with the portfolio UI: hero, skills, projects, architecture, terminal demo and contact.

Browser\n |\n index.html\n |\n Website

Why / use case: Why: HTML provides the webpage structure that GitHub Pages can serve.

## STEP 3 — Commit

We committed the initial portfolio code to the main branch.

Changes\n \ncommit\n \nGitHub/main

Why / use case: Why: a commit creates a versioned checkpoint. This is the same basic Git workflow used before CI/CD.

## STEP 4 — GitHub Pages

Settings → Pages → Deploy from a branch → main → /(root).

main\n \n/(root)\n \nGitHub Pages

Why / use case: Why: this turns repository files into a hosted static website.

## STEP 5 — Temporary URL

GitHub published the site first at <https://ashwanishahi155.github.io/ashwani-devops-portfolio/>.

Code OK ✓\nPages OK ✓\nWebsite renders ✓

Why / use case: Why: we verified hosting and HTML rendering before introducing DNS. This isolates problems layer by layer.

## STEP 6 — Custom Domain

In GitHub Pages we entered ashwaniai.site.

ashwaniai.site\n \nGitHub Pages

Why / use case: Why: a branded domain is more professional than a project URL.

## STEP 7 — GoDaddy DNS

We opened the DNS records for the domain. An old root A record pointed to WebsiteBuilder Site.

Before:\nashwaniai.site → WebsiteBuilder\n\nAfter:\nashwaniai.site → GitHub Pages

Why / use case: Why: the domain must point to the service actually hosting the website.

## STEP 8 — Four A Records

We replaced the root destination with GitHub Pages' documented apex-domain A records.

A @ 185.199.108.153\nA @ 185.199.109.153\nA @ 185.199.110.153\nA @ 185.199.111.153

Why / use case: Why: A records map a hostname to IPv4 addresses. These four addresses are documented by GitHub Pages; they are not our Azure/AKS IPs.

## STEP 9 — What is @?

In DNS, @ means the apex/root domain. Here, @ = ashwaniai.site.

@\n↓\nashwaniai.site

Why / use case: Why: it distinguishes the root domain from subdomains such as www, axion-ui and telemetry.

## STEP 10 — www CNAME

We configured CNAME www → ashwanishahi155.github.io.

www.ashwaniai.site\n |\n CNAME\n |\n v\nashwanishahi155.github.io

Why / use case: Why: www is a subdomain; GitHub documents the CNAME-to-GitHub-Pages-hostname pattern.

## STEP 11 — Preserve Existing Services

We did not change axion-ui → 51.8.208.43 or telemetry → 51.8.208.43.

ashwaniai.site → GitHub Pages\naxion-ui.ashwaniai.site → existing service\ntelemetry.ashwaniai.site → existing service

Why / use case: Why: a DNS zone can host multiple services. We changed only records required for the portfolio.

## STEP 12 — DNS Validation

GitHub initially showed a DNS-check warning, then later showed DNS valid for primary.

ashwaniai.site\n |\n DNS lookup\n |\nGitHub Pages\n |\nDNS valid ✓

Why / use case: Why: GitHub must verify that the custom domain resolves to GitHub Pages.

## STEP 13 — TLS

GitHub showed TLS certificate is being provisioned.

ashwaniai.site\n |\n TLS cert\n |\nhttps://ashwaniai.site □

Why / use case: Why: the certificate enables secure HTTPS access to the custom domain.

### 3. The Four GitHub Pages IPs — Deep Explanation

These are GitHub Pages' documented IPv4 addresses for apex/root custom-domain configuration. We did not invent them. They are not our Azure, AKS, database or GoDaddy server IPs.

185.199.108.153  
185.199.109.153  
185.199.110.153  
185.199.111.153

A record meaning: an A record maps a hostname to an IPv4 address. For this project, @ = ashwaniai.site.



Important: Do not think of the four IPs as frontend/backend/database/Kubernetes. They are the GitHub-documented GitHub Pages addresses used for apex-domain DNS configuration.

### 4. Final DNS Design

| Type  | Name      | Value                     | Purpose            |
|-------|-----------|---------------------------|--------------------|
| A     | @         | 185.199.108.153           | GitHub Pages apex  |
| A     | @         | 185.199.109.153           | GitHub Pages apex  |
| A     | @         | 185.199.110.153           | GitHub Pages apex  |
| A     | @         | 185.199.111.153           | GitHub Pages apex  |
| CNAME | www       | ashwanishahi155.github.io | www → GitHub Pages |
| A     | axion-ui  | 51.8.208.43               | Existing service   |
| A     | telemetry | 51.8.208.43               | Existing service   |

### 5. Real-World Recruiter Use Case



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v  
Ashwani Shahi  
Senior Azure DevOps Engineer  
|  
+-- Azure  
+-- Terraform  
+-- Kubernetes / AKS  
+-- CI/CD  
+-- DevSecOps  
+-- Monitoring / LGTM  
+-- Projects  
+-- Architecture

Business value: instead of sending only a PDF resume, the candidate provides a live, branded technical portfolio demonstrating Git, hosting, DNS and HTTPS knowledge through a working project.

## 6. Troubleshooting Journey

| Observation                          | Meaning / action                                                                                           |
|--------------------------------------|------------------------------------------------------------------------------------------------------------|
| DNS check unsuccessful               | DNS was not yet fully recognized/propagated; verify records and wait rather than randomly changing records |
| DNS valid for primary                | Root domain DNS was recognized correctly.                                                                  |
| TLS certificate is being provisioned | GitHub was preparing HTTPS certificate for the custom domain.                                              |
| www not valid initially              | Verify CNAME www → ashwanishahi155.github.io and allow DNS propagation.                                    |

## 7. Interview Answer — 30 Seconds

“I created a public GitHub repository and committed my static portfolio website to the main branch. I enabled GitHub Pages and deployed from the root of the main branch. After validating the default GitHub Pages URL, I mapped my GoDaddy domain as a custom domain. For the apex domain, I configured the four official GitHub Pages A records, and for the www subdomain I configured a CNAME pointing to my GitHub Pages hostname. GitHub then validated the DNS configuration and started provisioning the TLS certificate, allowing the portfolio to be served securely over HTTPS.”

## 8. One-Page Cheat Sheet

### GOAL

Make <https://ashwaniai.site> serve a professional DevOps portfolio without buying GoDaddy Website Builder.

### FLOW

CODE → GITHUB → COMMIT → GITHUB PAGES  
→ CUSTOM DOMAIN → GODADDY DNS  
→ A RECORDS + CNAME → DNS VALIDATION  
→ TLS → HTTPS → LIVE PORTFOLIO

### A RECORDS

A @ 185.199.108.153  
A @ 185.199.109.153  
A @ 185.199.110.153  
A @ 185.199.111.153

### CNAME

www → ashwanishahi155.github.io

### KEY TERMS

@ = root/apex domain = ashwaniai.site  
A = hostname → IPv4  
CNAME = hostname → hostname

GoDaddy = domain + DNS  
GitHub = repository + Pages hosting  
TLS = HTTPS security

## 9. Official Documentation

GitHub's official custom-domain documentation lists the four apex-domain A records and explains the www CNAME pattern. GitHub's HTTPS documentation covers TLS/HTTPS for custom domains.

Reference: [GitHub Docs — Managing a custom domain for your GitHub Pages site](#); [GitHub Docs — Securing your GitHub Pages site with HTTPS](#).